

02.07.2026 MACHINE LEARNING DEGREE PRESENTATION

Aesthetic Emotion Classification

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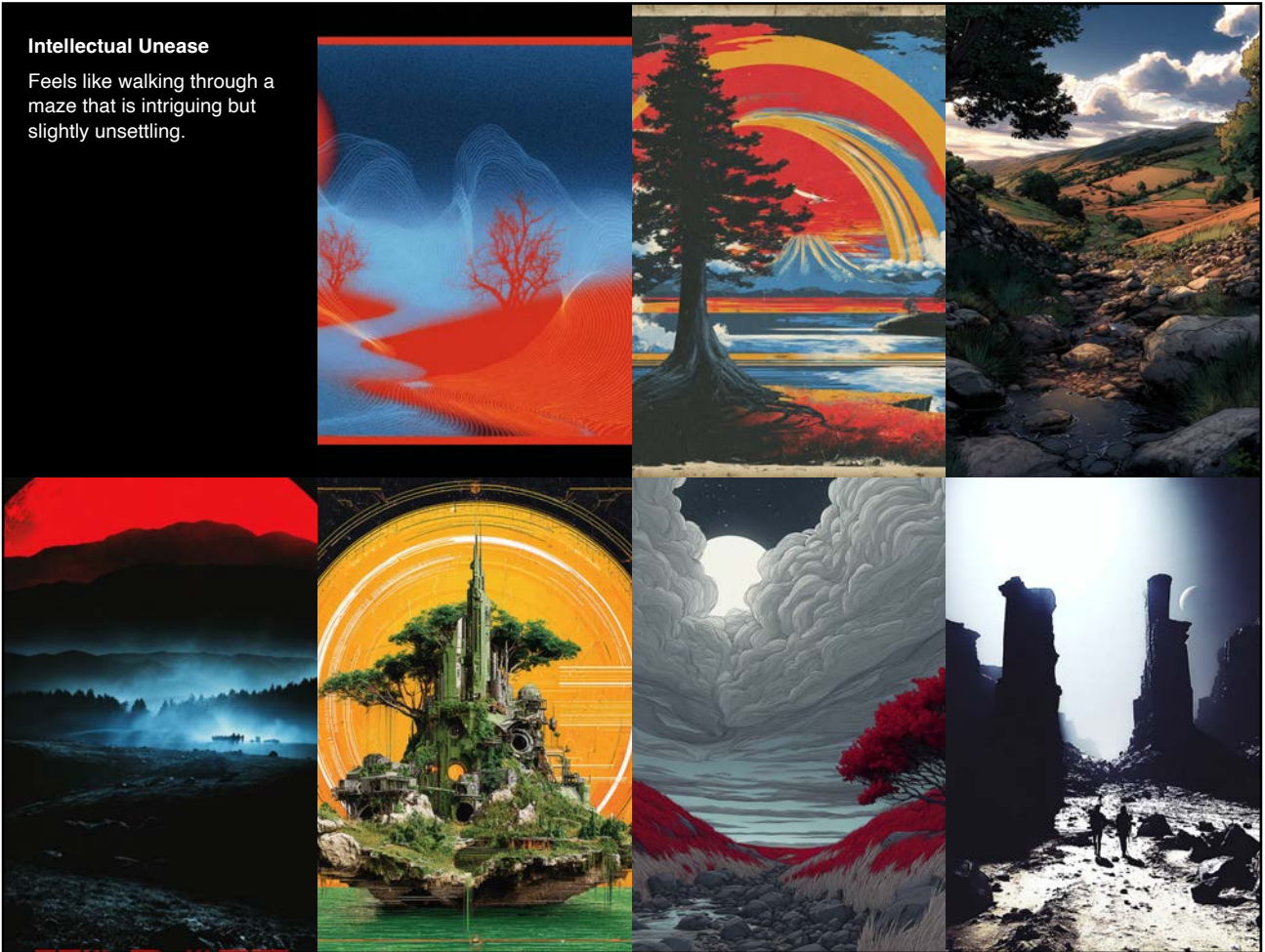
Introduction

Our project aims to predict the emotions evoked by visual styles in images using machine learning.

But before we look at training a model let's see if we can train you for the task.

Intellectual Unease

Feels like walking through a maze that is intriguing but slightly unsettling.



Here you see samples from the "Intellectual Unease" class.

The styles should feel like "walking through a maze that is intriguing but slightly unsettling".

Do you feel the uneasiness?

Melancholic

Feels like finding an old photograph and remembering a time that is gone forever.



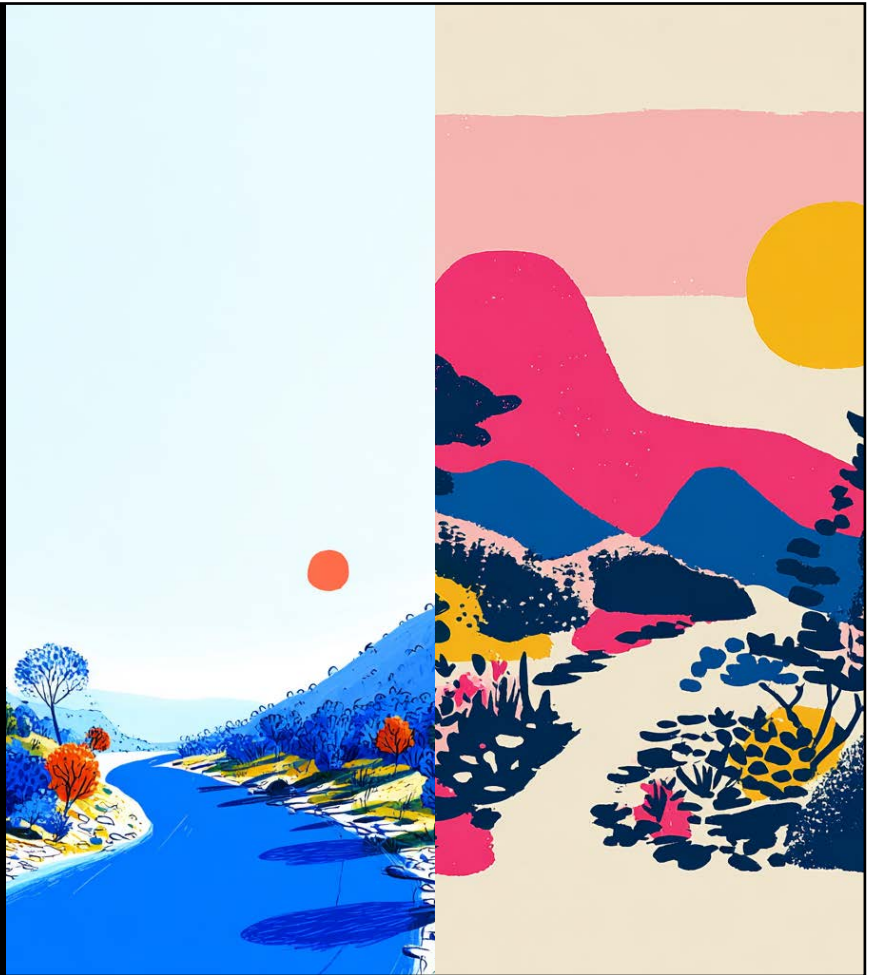
Here are samples from the "Melancholic" class.

The styles should feel like "finding an old photograph and remembering a time that is gone forever".

Do you feel the sadness?

Energetic Playfulness

Feels like dancing when your favorite song unexpectedly comes on.

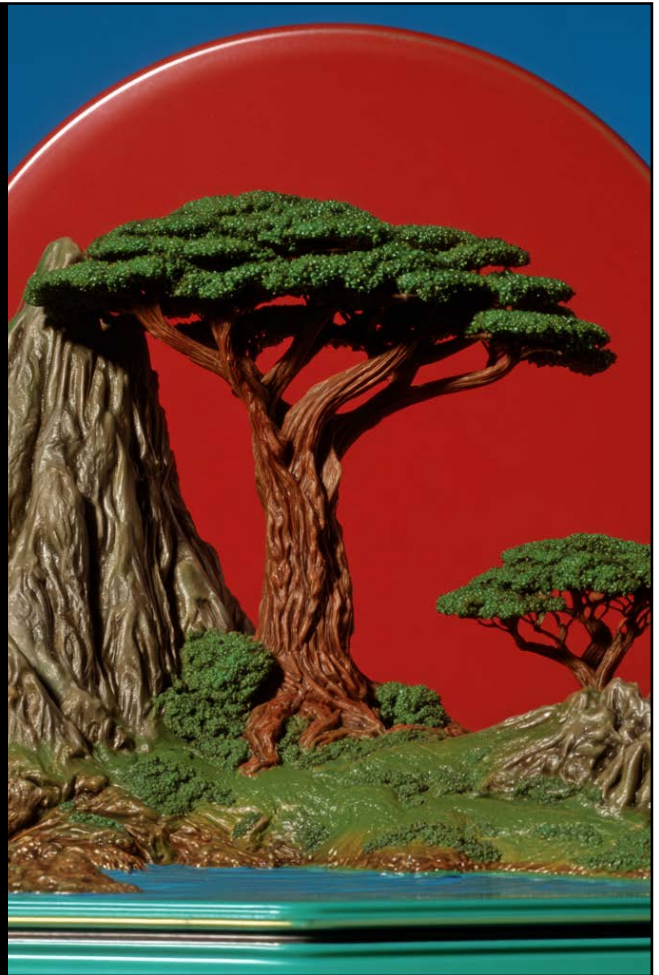


These are samples from the "Energetic Playfulness" class.

The styles should feel like "dancing when your favorite song unexpectedly comes on".

Do you feel energized?

Can we train a machine to predict emotion categories solely based on visual features?



Let's put you to the test.

Which class would you assign this image to?

Intellectual Unease, Melancholic or Energetic Playfulness?

Not easy, right? According to our dataset it is "Energetic Playfulness".

We chose this ambiguous style on purpose to show how difficult and somewhat subjective the task is.

So the question remains, can we train a machine to predict emotion categories solely based on visual features?

Literature Review

Large-Scale Image Emotion Recognition: Fine-tuned CNNs trained on strongly labeled emotion images significantly outperform handcrafted features and generic ImageNet models across eight emotion categories.¹

ArtEmis: Links 80,000 artworks with 450,000 emotional reactions and explanations, showing that subjective responses to art still contain learnable patterns.²

Learning from Web Data: EmotionNet uses over one million emotion-tagged stock images to learn affective visual features that generalize better than standard ImageNet features.³

1. <https://cdn.aaai.org/ojs/9987/9987-13-13515-1-2-20201228.pdf> ↗

2. https://openaccess.thecvf.com/content/CVPR2021/papers/Achlioptas_ArtEmis_Affective_Language_for_Visual_Art_CVPR_2021_paper.pdf ↗

3. https://openaccess.thecvf.com/content_CVPR_2020/papers/Wei_Learning_Visual_Emotion_Representations_From_Web_Data_CVPR_2020_paper.pdf ↗

The literature supports three decisions in this project: use CNNs as the core model family, evaluate carefully because emotion labels are subjective, and test transfer learning instead of relying only on a small custom CNN.

Dataset Characteristics

6,832 AI-generated RGB images at 224 × 224 pixels

427 styles × 4 motifs × 4 variations

Labels: 8 Emotion Categories

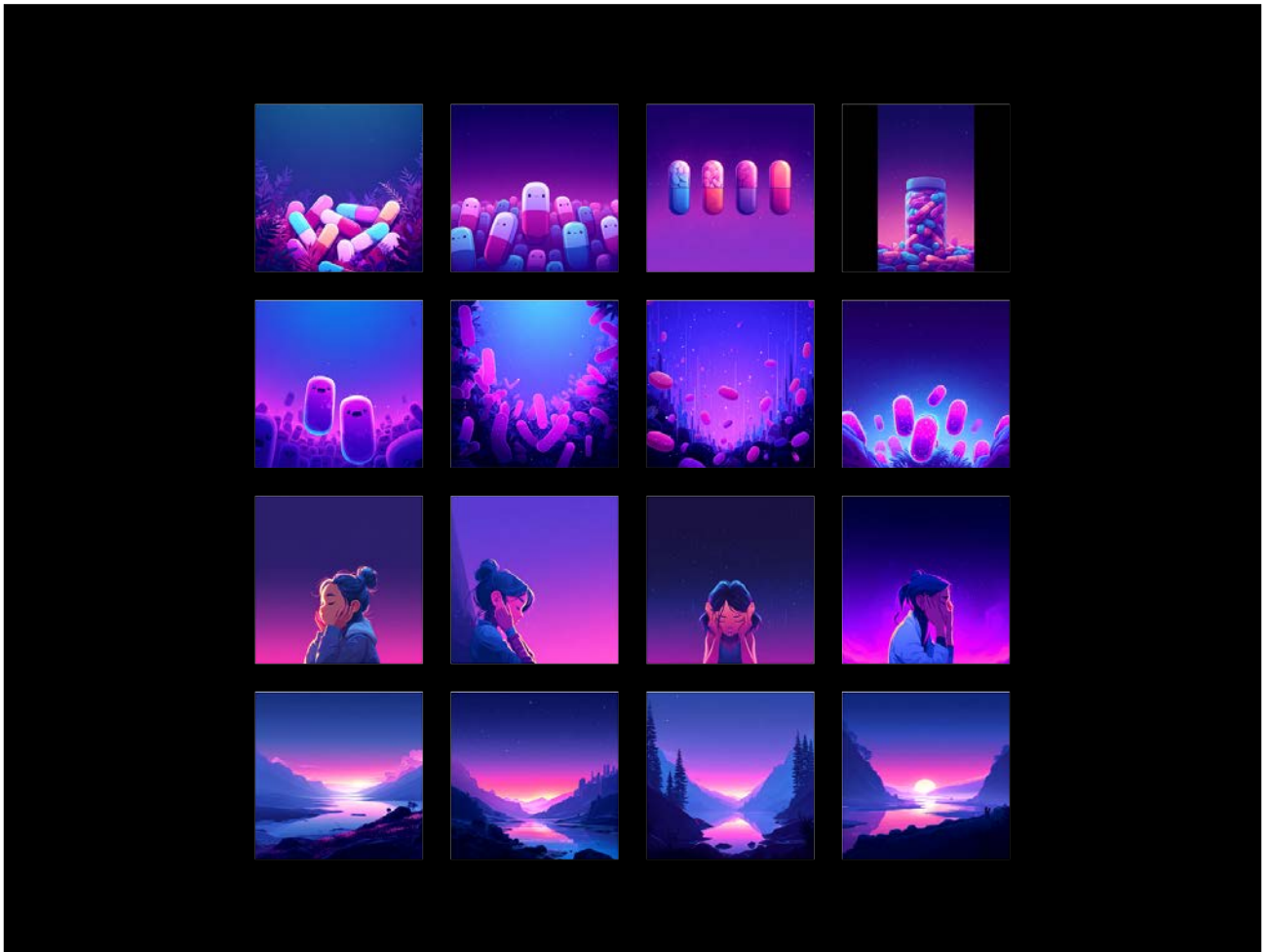
Style-level split train/validation/test (70/15/15)

Labels derived from: cluster analysis of LLM emotion annotations

In a prior study we generated 8000 images using midjourney by only using the motif + style code e.g.
`bacteria --sref 4255112626`

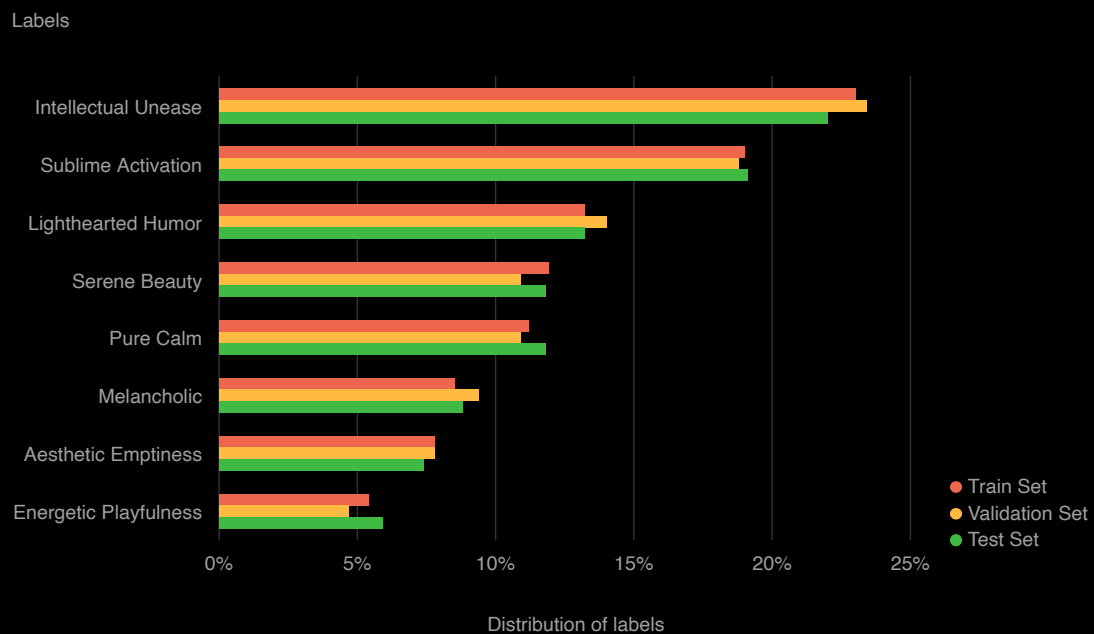
We then let an LLM annotate the images using the AESTHEMOS questionnaire to get an emotion profile per image

After that we performed cluster analysis on those 21 dimensional profiles to get 8 clusters that form our labels



Here you can see all sixteen images for one style

The Rows indicate the motifs while the columns indicate the variation

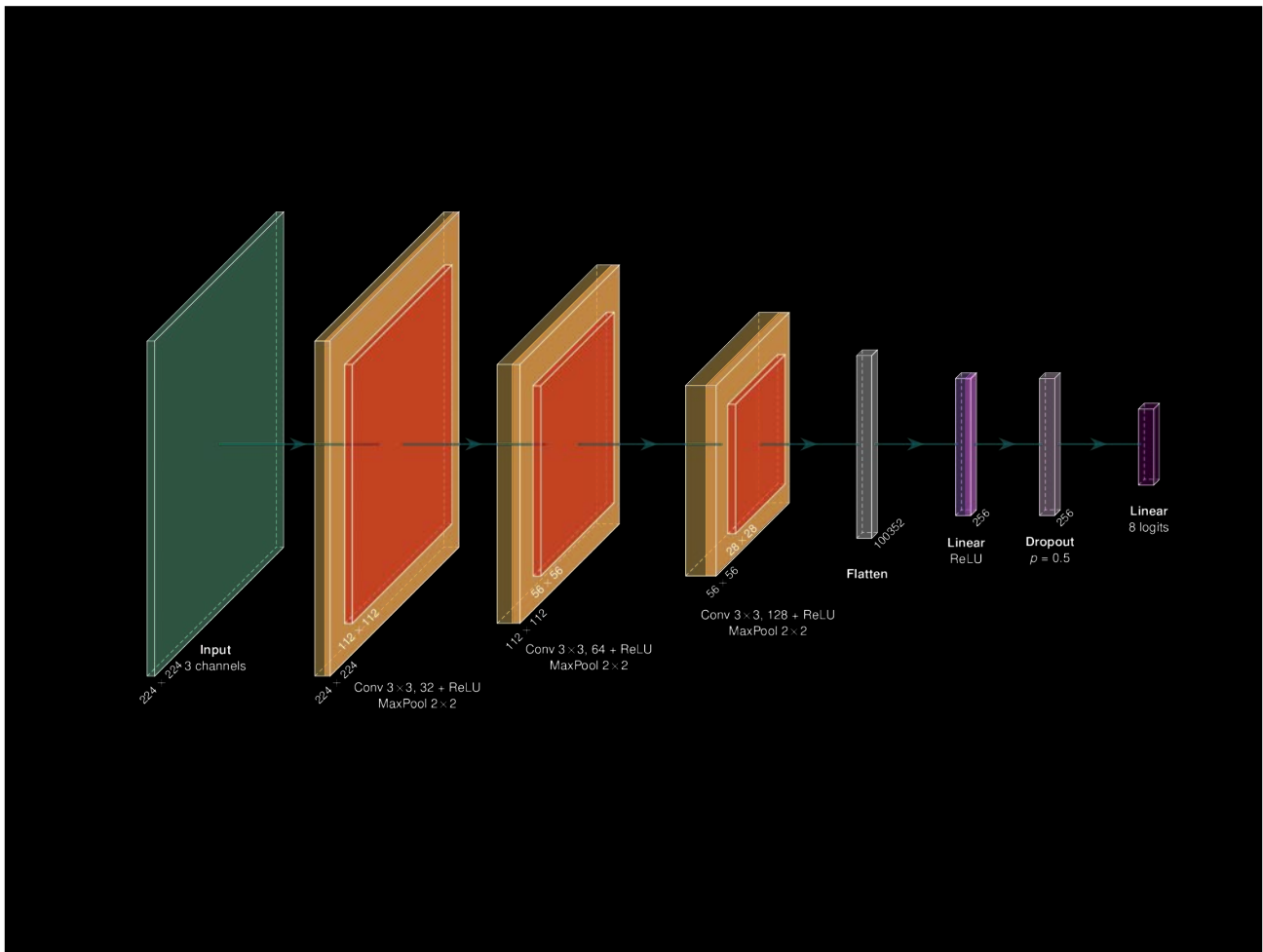


Bar Chart illustrating the label imbalance

The dataset is highly imbalanced: most images belong to the intellectual unease class while the least images belong to the energetic playfulness class.

We tried to retain the relative distribution across train set, validation set and test set.

Baseline Model



Our baseline is a simple CNN trained from scratch.

It has three convolutional layers with ReLU activation and max-pooling, using 32, 64, and 128 feature maps.

We use three-by-three kernels throughout the network, followed by a fully connected classifier with dropout.

The model was trained for ten epochs using the Adam optimizer.

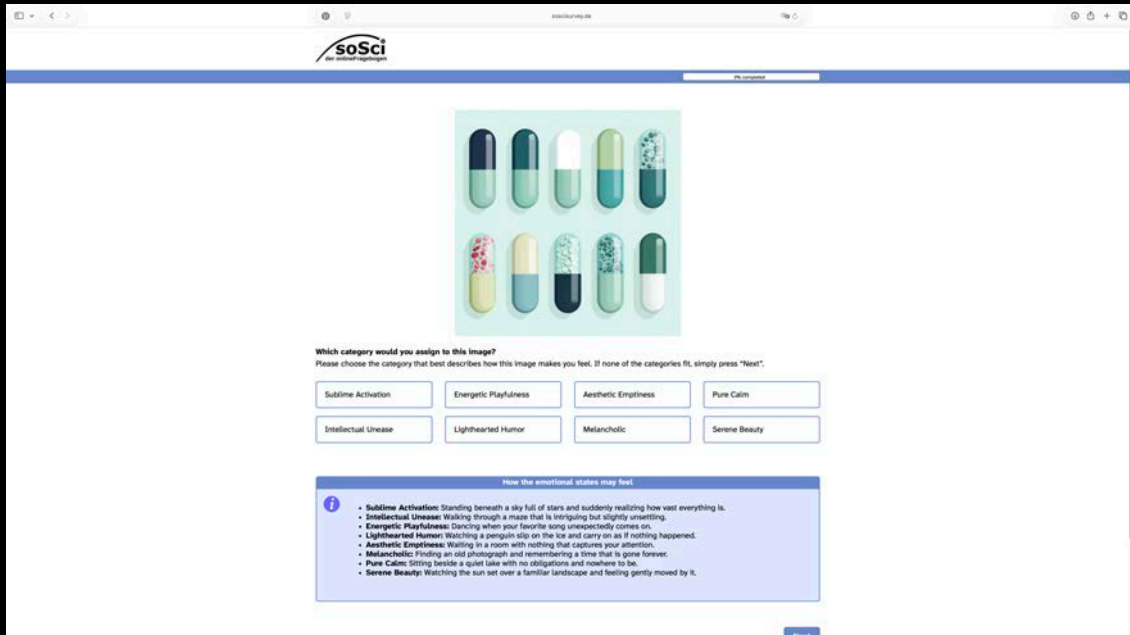
Baseline Model

Metric	Score
Accuracy	0.3998
Weighted precision	0.3883
Weighted recall	0.3998
Weighted F1	0.3748
Loss	2.9650

Although the model achieved high training accuracy, it failed to generalize to unseen data, indicating overfitting.

Still, it performed better than random guessing, which would achieve an accuracy of 12.5%.

Model Definition and Evaluation



To put the model's performance into perspective, we estimated human performance on the same task.

We created a survey with 32 images—four per emotion cluster across two motifs—and asked five participants to classify them.

Human Baseline Estimation (N = 5)

Metric	Mean	Std. deviation	Median
Accuracy	0.3375	0.0559	0.3750
Precision	0.3431	0.0569	0.3241
Recall	0.3375	0.0559	0.3750
F-Score	0.3868	0.0481	0.3898

Even humans struggled to classify the eight abstract emotion clusters, showing that this is a subjective and challenging task.

The human survey provides context for task difficulty, but it is not a direct full-test-set benchmark.

Hyperparameter Optimization with k-fold Cross Validation

Item	Value	Hyperparameter	Search space	Best value	Tuning metric	score
Method	Optuna	n layers	1–3	3	Weighted F1 (5-fold CV)	0.32
Objective	Maximize mean weighted F1-score	n filters layer 0	8–64, step 8	48		
Evaluation	5-fold cross-validation	n filters layer 1	8–64, step 8	32		
Metric	Average weighted F1 across folds	n filters layer 2	8–64, step 8	48		
Number of trials	20	kernel size layer 5	[3, 5]	5		
		kernel size layer 5	[3, 5]	5		
		kernel size layer 5	[3, 5]	5		
		Dropout	0.1–0.5	0.42		
		fc size	64–512, step 64	128		
		Learning rate	1e-4–1e-2	1e-3		
		Dropout	0.1–0.5	0.4		
		Resolution	[16, 32, 64]	64		
		Batch Size	[8, 16, 32, 64]	16		

We then optimized the baseline model using Optuna with five-fold cross-validation.

We also replaced the standard loss with a weighted loss function to compensate for class imbalance.

Despite tuning the hyperparameters, performance was slightly worse than the original baseline.

```

def update_model_last_layer(model, num_classes):
    """
    Freezes the feature layers of a pre-trained model and replaces its final
    classification layer with a new one adapted to the specified number of classes.

    Args:
        model (torch.nn.Module): The pre-trained model to be modified.
        num_classes (int): The number of output classes for the new classification layer.

    Returns:
        torch.nn.Module: The modified model with frozen feature layers and a new
        classification layer.
    """

    for name, param in model.named_parameters():
        if not name.startswith("fc."):
            param.requires_grad = False

    last_classifier_layer = model.fc

    num_features = last_classifier_layer.in_features

    new_classifier = nn.Linear(in_features=num_features, out_features=num_classes)

    # Replace the original last classification layer with the newly created layer
    model.fc = new_classifier

    return model


def create_model():
    weights = ResNet50_Weights.DEFAULT
    model = resnet50(weights=weights)
    model = update_model_last_layer(model, num_classes=8)
    return model.to(DEVICE)

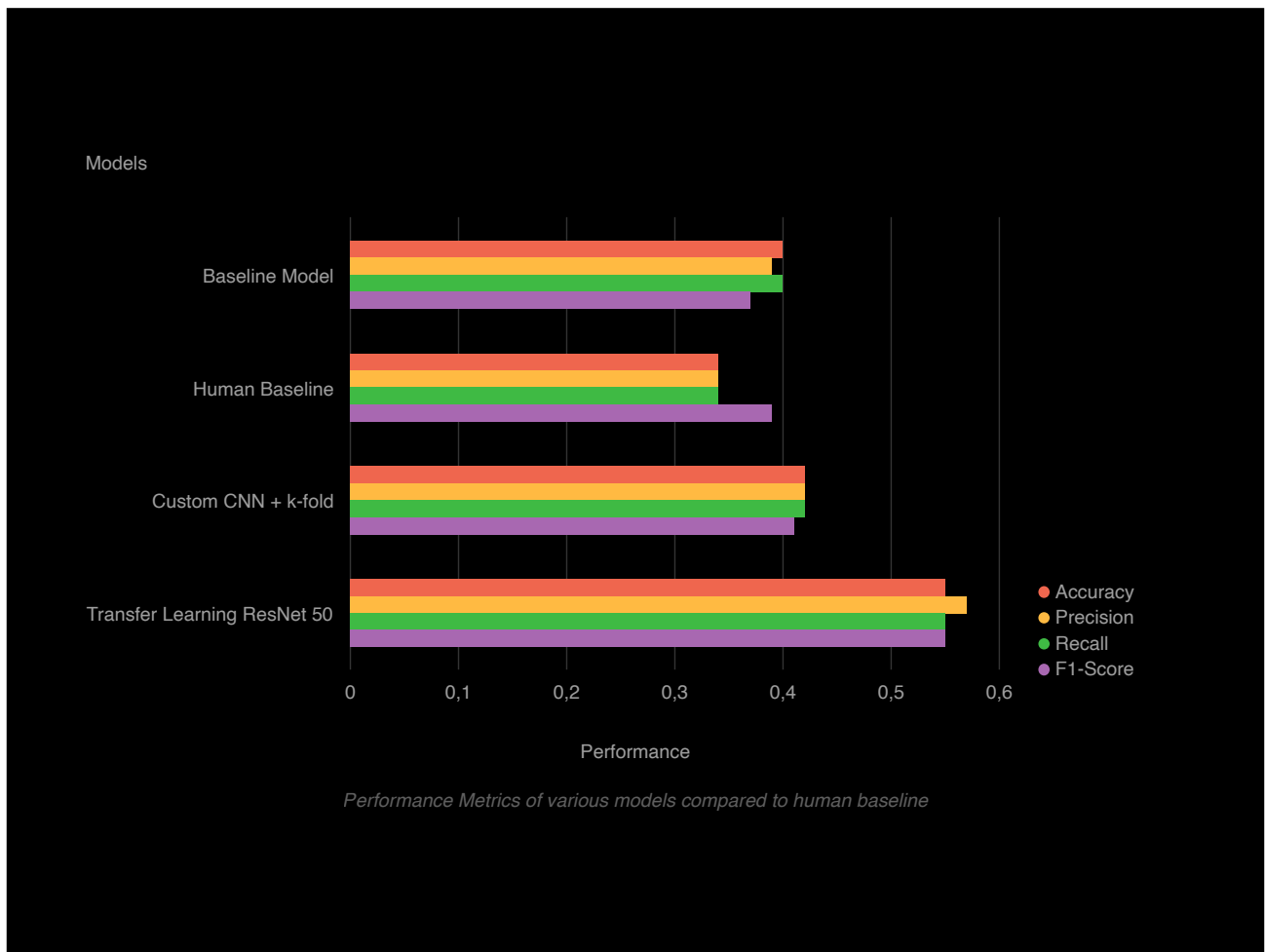
```

We therefore turned to transfer learning.

We selected ResNet50, a widely used model pretrained on over one million ImageNet images.

We froze the pretrained feature extractor and replaced the final classification layer to predict our eight emotion clusters.

Results



As you can see, the transfer learning model clearly outperformed all previous approaches, achieving the best performance across every metric.

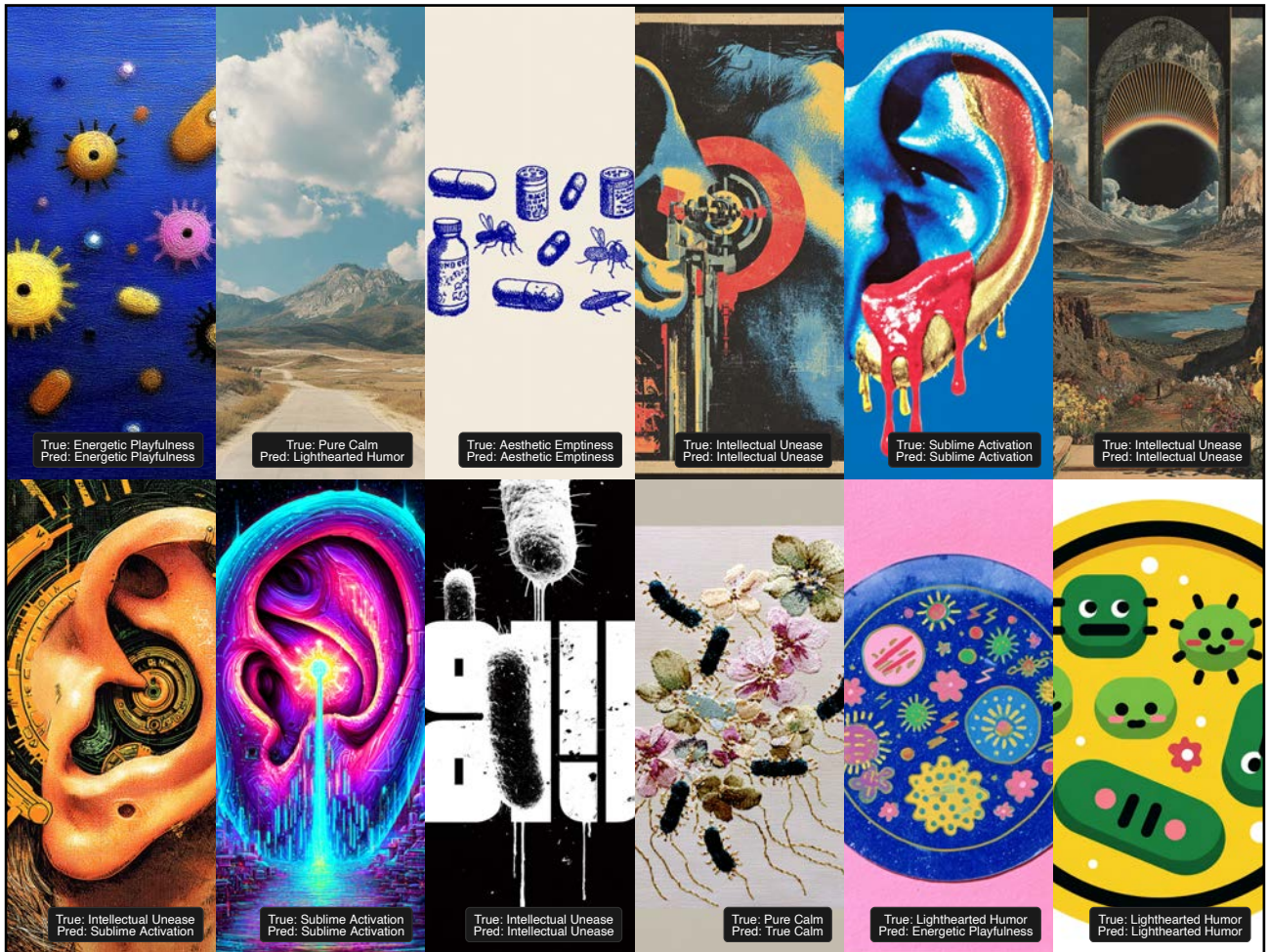
True Label	Aesthetic Emptiness	41%	0%	9%	3%	13%	25%	9%	1%
	Energetic Playfulness	3%	30%	6%	23%	3%	13%	6%	16%
	Intellectual Unease	5%	1%	63%	0%	7%	1%	9%	14%
	Lighthearted Humor	8%	16%	1%	65%	0%	6%	4%	0%
	Melancholic	4%	1%	26%	2%	51%	6%	6%	3%
	Pure Calm	5%	0%	5%	8%	14%	55%	13%	0%
	Serene Beauty	1%	2%	8%	5%	5%	23%	48%	8%
	Sublime Activation	1%	14%	11%	0%	0%	5%	15%	54%
		Aesthetic Emptiness Energetic Playfulness Intellectual Unease Lighthearted Humor Melancholic Pure Calm Serene Beauty Sublime Activation							
		Predicted Label							

This confusion matrix shows the model's predictions for each class.

The y-axis represents the true labels, and the x-axis the predicted labels. A strong diagonal indicates good performance.

The model classified Intellectual Unease reliably but struggled with Energetic Playfulness.

It frequently confused Energetic Playfulness with Lighthearted Humor, as well as Melancholic with Intellectual Unease and Pure Calm with Serene Beauty.



Here are twelve randomly selected test images with their ground-truth labels and the model's predictions.

The model classified eight correctly and misclassified four.

With that, I'll hand over to Cem.

Challenges and Errors

- Heterogeneous visual styles within classes
- Gradual and overlapping class boundaries
- LLM-generated annotations lack human validation
- Potential mismatch between initial and articulated human emotions
- Preventing style leakage across data splits
- Handling imbalanced emotion classes
- Comparing model performance against a small human baseline
- Avoiding overfitting on a limited image dataset

Discussion

- More training data could improve performance
- Emotion labels are inherently subjective
- Training may be sensitive to random initialization (seed)

Conclusion and Future Work

- Model outperforms the human baseline (small sample)
- Pretrained visual features outperform CNNs trained from scratch
- Current model is not yet sufficiently reliable
- Feature map analysis could provide additional insights
- Personalized emotion prediction

Q&A